Journal of Experimental Marine Biology and Ecology 443 (2013) 75–84

Contents lists available at SciVerse ScienceDirect

Journal of Experimental Marine Biology and Ecology

journal homepage: www.elsevier.com/locate/jembe

Observer bias and detection probability in underwater visual census of fish assemblages measured with independent double-observers

A.T.F. Bernard [a,b,*], A. Götz [b,e], S.E. Kerwath [c,d], C.G. Wilke [c]

[a] Zoology and Entomology Department, Rhodes University, Grahamstown, 6139, South Africa
[b] Elwandle Node, South African Environmental Observation Network (SAEON), Grahamstown, 6139, South Africa
[c] Line-Fish Research Division, Department of Agriculture, Forestry, and Fisheries, Cape Town, South Africa
[d] Zoology Department, University of Cape Town, Cape Town, South Africa
[e] Zoology Department, Nelson Mandela Metropolitan University, Port Elizabeth, 6031, South Africa

ARTICLE INFO

Article history:
Received 19 August 2012
Received in revised form 20 January 2013
Accepted 24 February 2013
Available online 20 March 2013

Keywords:
Double-observer
Observer bias
Rare species
Species detection
UVC
Volunteers

ABSTRACT

Underwater visual census (UVC) is one of the more frequently used methods for monitoring of subtidal fish communities. However, population estimates from UVC are plagued by numerous biases that ultimately reduce the diagnostic power of the data. Adaptations to methods that account for detection errors by employing multiple observers can reduce many of these biases, but have received little consideration in past marine research. In the terrestrial environment the double-observer method has been successfully applied to estimate visibility bias, observer bias and detection probabilities. We used an independent double-observer method to measure the influence of observer type (volunteers and researchers) and fish community structure on the detection probability and count dissimilarity between two observers simultaneously surveying the same transect. Community data collected by volunteers were of a consistently lower quality than that collected by researchers. However, this was not the case for the dominant species for which researchers and volunteers produced similar data. The analysis further suggested that rare species contributed disproportionately to the bias in UVC data because the abundance of a species within a survey area determined its detection probability. Our results highlight the need to account for detection probability during UVC and it is recommended that future research based on UVC integrates an independent double-observer approach. Furthermore, the results indicate that care needs to be taken when designing monitoring programmes as allowing inexperienced volunteers to participate can undermine the integrity of the data.
© 2013 Elsevier B.V. All rights reserved.

1. Introduction

Underwater visual census (UVC) is widely used to assess the status of fish populations in relation to climate change, resource exploitation and management actions (Edgar et al., 1997; Harborne et al., 2008; Jackson et al., 2001; McClanahan et al., 2007; Ward-Paige et al., 2010). However, it is generally accepted that this method is plagued by biases from multiple sources. With UVC strip transects, bias can originate from transect length (Kulbicki et al., 2010), width (Cheal and Thompson, 1997), and the swimming speed of the observer (Lincoln-Smith, 1988). Observer related bias is typically large, as individual observers differ in their ability to accurately estimate distance underwater (Harvey et al., 2004; Thresher and Gunn, 1986), correctly identify species (Thompson and Mapstone, 1997) and estimate the size of individuals (Edgar et al., 2004). Observer experience also influences the precision of data collected with UVC (Thompson and Mapstone, 1997; Williams et al., 2006). Furthermore, environmental characteristics such as water clarity (MacNeil et al., 2008a, b) and habitat complexity (Edgar and Barrett, 1999) often affect the confidence around population estimates, while species abundance, size, appearance and behaviour alter their detectability (Bozec et al., 2011; Edgar et al., 2004; Kulbicki, 1998; MacNeil et al., 2008a, b; Willis, 2001). As a result UVC data reflect a complex mix of errors all of which increase the deviation in the data and mask the natural variability of the population.

The cost of establishing subtidal long-term monitoring programmes is considerable and often limits the ability to collect adequate data (Vos et al., 2000). This has led to increased interest in citizen-science initiatives that rely on recreational SCUBA divers to conduct UVC surveys (Ward-Paige et al., 2010). By allowing volunteers to participate the workforce is larger, the cost is reduced, and stakeholder participation is facilitated (Darwall and Dulvy, 1996; Edgar and Stuart-Smith, 2009; Halusky et al., 1994; Pattengill-Semmens and Semmens, 2003). A number of studies have shown that data collected by volunteers are of a similar quality to that collected by researchers, suggesting that citizen-science initiatives can contribute to monitoring objectives (Darwall and Dulvy, 1996; Edgar and Stuart-Smith, 2009; Mumby et al., 1995).

Significant improvements in population sampling methods are those that consider detection errors, include distance sampling and employ multiple-observer approaches (Elphick, 2008). Recent adaptations to the UVC transect method have begun to alleviate some of

* Corresponding author at: Zoology and Entomology Department, Rhodes University, Grahamstown, 6139, South Africa. Tel.: +27 46 622 9899.
E-mail address: ant@saeon.ac.za (A.T.F. Bernard).

0022-0981/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.jembe.2013.02.039

the underlying biases. For example, distance sampling (Bozec et al., 2011; Kulbicki, 1998, Kulbicki et al., 2010), and capture–mark–recapture (CMR) models based on repeated sampling of fixed strip transects (MacNeil et al., 2008a, b), both allow species-specific detection probabilities to be estimated. An alternative approach is to use double-observers where detection probabilities are measured from simultaneous surveys of the same sample using CMR models (Cook and Jacobsen, 1979; Jenkins and Manly, 2008). Most UVC surveys involve two observers, either with both conducting different surveys on adjacent sides of the transect line (e.g. Kulbicki, 1998), or where only one observer conducts the survey and the other serves as a buddy diver (e.g. Bennett et al., 2009). To date, no studies have investigated observer bias and species-specific detection probability with two observers simultaneously surveying the same strip transect.

Our aim was to develop and test an independent double-observer approach to traditional UVC strip transects that would allow direct comparison of data collected by two observers. We then employed this method to measure the influence of observer type (researcher or volunteer), environmental characteristics and fish community structure on the detection probability and count dissimilarity when looking at the entire fish community and individually for the dominant fish species.

2. Materials and methods

2.1. Study area

Two marine protected areas (MPAs) within the Agulhas Ecoregion of South Africa were selected as study areas. The False Bay section of the Table Mountain National Park (TMNP) MPA lies on the western edge of the Agulhas Ecoregion (Fig. 1a). The study area was the Castle Rock no-take zone (NTZ) and the adjacent Caravan Reef (Fig. 1b) within the TMNP MPA. The Castle Rock NTZ is a small (6 km^2), but old (established in 1979) marine reserve where all types of extractive use are prohibited. Caravan Reef is an extension of the reef complex within the Castle Rock NTZ (Fig. 1c), however hook and line and spear fishing are permitted on this reef. The Tsitsikamma National Park (TNP) MPA lies in the centre of the Agulhas Ecoregion (Fig. 1a). It is the oldest (established in 1964) and one of the largest (323 km^2) no-take MPAs in Africa. The study area was situated to the east of Storms River mouth in the centre of the TNP MPA (Fig. 1d) and consisted of a diverse reef complex of high and low profile rocky reef and intermittent sand patches (Fig. 1e). Both study areas were stratified according to depth and reef profile. Depth was classified as shallow (5–18 m) and deep (18–25 m), while reef profile consisted of low and high profile bathymetry. Sandy bottom, or sand inundated reefs were avoided. Sample stations were evenly distributed between the different strata.

2.2. Paired-transects

The sampling method developed consisted of a combination of an independent double-observer (IDO) transect (Jenkins and Manly, 2008) with a single-observer UVC transect (Bennett et al., 2009). In the traditional IDO method, two observers conduct independent surveys of the same area at the same time recording the geographic position of each sighting along a transect (Jenkins and Manly, 2008). This allows direct comparison of the data collected by two observers and facilitates the calculation of detection probabilities for unique individuals or groups. Underwater visual census strip transects can be complicated as the density of fish is often very high, and they move rapidly through the survey area. As such it was felt that the IDO approach would be too time consuming and complicated for inexperienced volunteers, and a simplified method that provided only total abundance of individuals per species, per observer, per transect was employed (referred to as paired-transects from this point forward).

For the paired-transects, two observers would initially lay a 50 m line along the reef in a randomly selected direction. They would then position themselves as close together as possible on either side of the line and simultaneously swim back along the line independently identifying and counting all fish seen within a quarter-sphere, with a three metre radius directed forward and to the sides of the line, thereby surveying the same area at the same time. Fish that entered from behind the observers were ignored to avoid recounting individuals. Observers had to maintain a swimming speed of approximately 6 m per minute and every effort had to be made not to stagger their positions relative to each other, or drift off the line. The output from a paired-transect consisted of two independently collected species abundance lists. This allowed for the direct assessment of the hypothesis that there was no difference between the two datasets from one paired-transect.

To assess the change in precision with increasing experience the observers were required to conduct multiple paired-transects. For practical reasons, no constraints were placed on whom the observers paired with, so long as they were from the same *Observer type* (researcher or volunteer). When observers with different levels of experience paired together, the experience level was taken to be that of the observer with the least number of paired-transects.

For the data analysis four assumptions were made: (i) observations by paired observers were independent (i.e. there was no communication between the observers); (ii) all fish within the transect area were equally visible to both observers (i.e. not obscured by the other observer or by hidden from one observer by reef structure); (iii) only fish within the bounds of the transect area were counted; and (iv), all fish were correctly identified.

2.3. Observer training and testing

The observers consisted of experienced researchers (between four and 20 years experience of conducing UVC of fish in the ecoregion) and newly-trained, inexperienced volunteers. All the volunteers were qualified recreational SCUBA divers with between one and 30 years of experience diving in the region. The volunteers were trained as part of a pilot study to test the feasibility of a volunteer monitoring programme for the TMNP MPA (Bernard and Götz, 2011). All volunteers attended one of seven standardised two-day training course where they were taught how to identify all the target fish species and how to perform paired-transects.

Before and after the training course the volunteer's identification skills were assessed with photo tests, and their identification and counting skills were assessed after the training course with a video test. The fish-community data recorded by each observer during the video test was converted to a measure of species detection probability (Eq. (1)) and a measure of paired-transect dissimilarity (Eq. (5)), by comparing the results from the tests to the actual fish community present in the video. For comparative purposes the researchers that participated in this study were tested as described above.

The accuracy of underwater distance estimates was determined in a swimming pool experiment. Divers had to gauge the distances to 15 conspicuous buoys while swimming down a dummy transect line. The buoys were suspended in the water column between 0.1 and 1.5 m from the bottom and were placed at random distances between 0.5 and 4 m on either side of a transect line. The tests were performed with both single-observers and paired-observers to determine how the paired-transect approach affected distance estimates, compared to the traditional single-observer approach. The accuracy of the estimates was measured against the actual distances, and expressed as a percentage deviance, centred on zero (i.e. zero = 100% accuracy). The statistical analysis of the test results was conducted with paired Student t-tests and non-parametric Mann–Whiney U tests in the R environment (version 2.13.0, R Development Core Team, 2011).

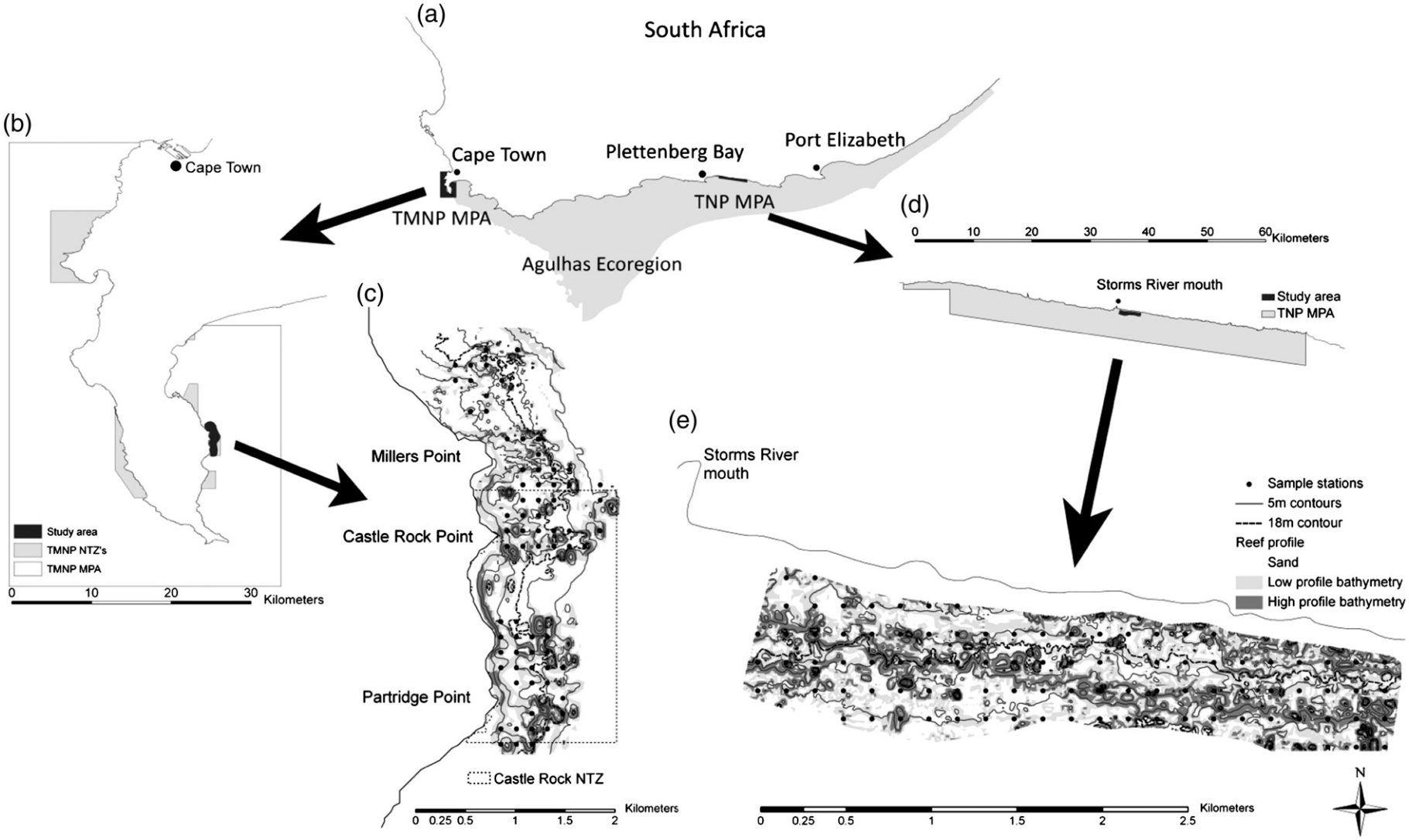

Fig. 1. Map of South Africa identifying the Agulhas Ecoregion (a), together with the location of the Table Mountain National Park (TMNP) marine protected area (MPA) with no-take zones (NTZs) (b) and the Tsitsikamma National Park (TNP) MPA (d). Detailed bathymetric maps of the Castle Rock (c) and Rheeders Reef (e) study areas and positions of potential sample stations are included.

Table 1
Description of the covariates included in the full generalised linear models.

Name	Description	Levels
Observer type	Measure of the background experience of the different observers	Researcher Volunteer
Experience level	Minimum number of replicate paired-transects conducted by the observers within a pair	1–12
Location	Study area	Castle Rock or Rheeders Reef
Reef profile	Reef bathymetry	High or low
Species richness	Total number of species recorded during a paired-transect	1–17
Average species abundance	The total number of fish recorded divided by the species richness per paired-transect	1–47
Abundance	The abundance of a individual species per paired-transect	1–125

2.4. Community level similarity indices

The precision of the count data collected during a paired-transect was estimated with a measure of species detection probability, the Jaccard coefficient (JC), and a measure of paired-transect dissimilarity, CY dissimilarity (CYd) (Cao et al., 1997). An alternative dissimilarity measure, the binomial deviance dissimilarity (Anderson and Millar, 2004), was explored, however CY dissimilarity was preferred as it is averaged by the number of species recorded per paired-transect. This accounts for variation in species richness between different paired-transects, and allows for accurate comparison of the dissimilarity scores. The species detection probability provided the odds that both observers would see all species along a paired-transect, and was defined as:

$$JC = \frac{B}{S_2 + S_2 + B}, \tag{1}$$

where species recorded only by observer one are labelled S_1, only by observer two S_2, and by both observers B. Generalised linear models (GLMs) were employed to assess the effect of fish community structure (*Species richness* and *Average species abundance*), environment (*Location* and *Reef profile*) and observer (*Observer type* and *Experience level*) on the species detection probability (Table 1). The data were modelled using a binomial GLM with the logit link according to the equation below:

$$JC_i \sim B(\pi_i, n_i) \tag{2}$$

$$E(JC_i) = \pi_i \times n_i \quad and \quad var(JC_i) = n_i \times \pi_i \times (1-\pi_i) \tag{3}$$

$$logit(\pi_i) = \alpha + \beta_1(Observer\,type) + \beta_2(Experience\,level) + \beta_3(Location) + \beta_4(Profile) + \beta_5(Species\,richness) + \beta_6(Average\,species\,abundance) + \varepsilon_i, \tag{4}$$

where the expected response, $E(JC_i)$, is assumed to be binomially distributed with a probability of π_i from n_i independent trials. For presence/absence data, $n_i = 1$, while for proportional data, n_i is the total number of trials. The expected mean and variance for JC_i are given by $\pi_i \times n_i$ and $n_i \times \pi_i \times (1 - \pi_i)$, respectively. Eq. (4) presents the full model, with the predictor function consisting of the intercept, α, the covariates included as model parameters, $\beta(\cdot)$ and the error term, ε_i.

The paired-transect dissimilarity, CYd, overcomes bias in weighting related to variation in species abundance associated with traditional dissimilarity or similarity measures, such as the Bray–Curtis measure and the Canberra metric (Cao et al., 1997). Paired-transect dissimilarity is a distance measure ranging from zero to infinity, with identical samples equal to zero, and is expressed as:

$$CYd = \frac{1}{N_l}\sum\left(\frac{\left(X_{ijl} + X_{kjl}\right)\log_{10}\left(\frac{X_{ijl}+X_{kjl}}{2}\right) - X_{ijl}\log_{10}X_{kjl} - X_{kjl}\log_{10}X_{ijl}}{X_{ijl} + X_{kjl}}\right), \tag{5}$$

where X_{ijl} and X_{kjl} are the numbers of individuals of species j recorded by observer i and observer k during paired-transect l, while N_l is the total number of species observed during paired-transect l. The nominator section determines the dissimilarity between the two observers, and the denominator section removes the effects of absolute species abundance. The paired-transect dissimilarity is then averaged to take into account the variability in species richness, $\frac{1}{N_l}$. To account for zeros in the count data, 0.1 was added to all X_{ijl} and X_{kjl} values. According to Cao et al. (1997) this step avoids mathematical paradox and enables the measure to respond to all significant variation sensitively and weight them with minimum bias. The paired-transect dissimilarity data were modelled with a GLM fitted with a Gaussian distribution and an identity link:

$$CYd_i \sim N\left(\mu_i, \sigma^2\right) \tag{6}$$

$$E(CYd_i) = \mu_i \quad \text{and} \quad var(CYd_i) = \sigma^2, \tag{7}$$

where the expected response variable, CYd_i, is assumed to be normally distributed with mean μ_i, and variance σ^2 (Zuur et al., 2009). The full model was:

$$E(CYd) = \alpha + \beta_1(Observer\,type) + \beta_2(Experience\,level) + \beta_3(Location) + \beta_4(Profile) + \beta_5(Species\,richness) + \beta_6(Average\,species\,abundance) + \varepsilon, \tag{8}$$

where the terms in the predictor function are equivalent to those described for Eq. (4) (Zuur et al., 2009).

2.5. Species level similarity indices

The detection probability (DP) for paired-transects was further assessed for the dominant species recorded during the study. Only species that were observed on more than 30 paired-transects were included in the analysis (Table 2). The data were structured as presence/absence, with a species marked as present when recorded by both

Table 2
Population statistics for the dominant species recorded during the paired-transects.

Family	Scientific name	Common name	N[a]	Mean[b]	SD[b]	Min	Max
Sparidae	*Chrysoblephus laticeps*	Roman	85	7.57	6.76	1	40
Cheilodactylidae	*Chirodactylus brachydactylus*	Twotone fingerfin	72	6.68	6.94	1	33
Sparidae	*Pachymetopon blochii*	Hottentot	64	9.73	10.30	1	50
Cheilodactylidae	*Cheilodactylus fasciatus*	Redfingers	56	2.34	1.56	1	13
Sparidae	*Boopsoidea inornata*	Fransmadam	53	19.07	20.73	1	100
Sparidae	*Spondyliosoma emarginatum*	Steentjie	37	12.63	21.17	1	125
Sparidae	*Pachymetopon aeneum*	Blue hottentot	32	9.15	8.06	1	34
Cheilodactylidae	*Cheilodactylus pixi*	Barred fingerfin	31	3.13	3.10	1	16

[a] Species sorted by the % of the total number of paired-transects where they were present (N).
[b] The mean and SD are calculated from only the paired-transects where the species was present.

observers in a pair, and absent when only recorded by one observer in the pair. Due to differences in species abundance, each species was modelled separately and the full model was standardised between species. The full binomial GLM was described as:

$$logit(\pi_i) = \alpha + \beta_1(Observer\,type) + \beta_2(Profile) + \beta_3(Abundance), \quad (9)$$

where the terms in the predictor function of the equation are equivalent to those described for Eq. (4).

2.6. Exploratory analysis and model selection

Detailed exploratory analysis was conducted following the approach of Zuur et al. (2010). Where colinearity was identified between covariates, the covariates with the highest variance inflation factor (VIF) were sequentially removed from the model until all the VIFs were less than the threshold of three. Colinearity is defined as the presence of correlation between covariates, and it is problematic as it confuses statistical analysis (Zuur et al., 2010). This resulted in the removal of *Location* from all the analyses, as it was correlated with *Species richness*. To ensure that the omission of *Location* did not bias the results through a site-level correlation, patterns in the most parsimonious model residuals were investigated for each location (see Supplement 3). Regression diagnostic tests and model selection were carried out according to the methods of Logan (2010), whereby the Akaike information criterion (AIC) for all possible combinations of covariates included in the full model was calculated and the combination with the lowest AIC score was considered to be the most parsimonious model. All analyses and plotting were conducted in the R environment (version 2.13.0, R Development Core Team, 2011).

3. Results

3.1. Observer training and assessment

Sixty-one volunteers were tested prior to and after training. The ability to identify species improved significantly between the first (50.82 ± 13.95%) and second (79.92 ± 14.46%) tests ($t = 18.5$, $p < 0.001$). Although the researchers scored 13% higher than the volunteers, there was no significant difference ($W = 188.5$, $p > 0.05$) between the marks obtained by the volunteers following training (79.92 ± 14.46%) and the researchers (93.50 ± 3.79%). However, there was considerably greater variability in the volunteer results (coefficient of variation = CV = SD/mean = 0.18) compared to researchers (CV = 0.04), suggesting that the researchers had better accuracy and precision with their identification skills. Results from the video test indicated that the researchers were significantly better than the volunteers at identifying the species present in the video (JC: Vol = 0.76 ± 0.13, Res = 0.95 ± 0.04; $W = 25$, $p < 0.01$), as well as counting the individuals of a species (CYd: Vol = 0.40 ± 0.24, Res = 0.10 ± 0.11; $W = 252$, $p < 0.01$).

Both, the single- and paired-observer methods underestimated distance underwater, however the error was significantly higher ($W = 26130$, $p < 0.001$) for the single- (−14.11 ± 26.44%) compared to the paired-observer (−3.65 ± 25.65%) method.

3.2. Paired-transect sampling effort and data description

A total of 103 paired-transects were conducted, 71 in the TMNP and 32 in the TNP MPA. Of these, 65 were conducted by a group of 28 volunteers and 38 by a group of seven researchers. Twelve repeat dives (representing *Experience levels*) were conducted by each *Observer type* with at least three replicates per *Experience level*.

A total of 32 species of fish were recorded during the experiment with the *Sparidae* family contributing 50% (Supplement 1). Only eight species were observed on more than 30 paired-transects, accounting for 70% of the 607 species observations (Table 2).

3.3. Community level similarity indices

Species detection probability averaged (±SD) 0.69 (±0.22). The most parsimonious model included the covariates *Observer type* and *Average species abundance*. *Experience level*, *Reef profile* and *Species richness* were omitted from the most parsimonious model (Supplement 2). Analysis of the residuals of the most parsimonious model showed no evidence of a site-level correlation, suggesting that *Location* did not bias the results (Supplement 3). Likelihood ratio tests using the step-wise regression showed that the effect of *Observer type* was marginally not significant ($X^2 = 3.46$, $p = 0.06$), with researchers having a higher probability (0.71 ± 0.03) of detecting all species along a transect than volunteers (0.66 ± 0.07) (Fig. 2a). The effect of *Average species abundance* was significant ($X^2 = 13.56$, $p < 0.001$), with increasing abundance increasing the probability that all species would be detected (Odds ratio ± SE = 0.042 ± 0.01) (Fig. 2b).

Paired-transect dissimilarity (CYd) ranged between zero and 1.8, with a mean of 0.32 (± 0.23). The most parsimonious model excluded the effect of *Reef profile* (Supplement 4). There was no evidence of a site-level correlation in the model residuals, suggesting that *Location* did not bias the results (Supplement 3). The likelihood ratio test identified that the effect of *Observer type* was significant ($X^2 = 4.42$, $p < 0.05$), with the predicted dissimilarity scores greater for volunteers (0.33 ± 0.06) compared to researchers (0.28 ± 0.06) (Fig. 3a). Although the covariate *Experience level* was included in the most parsimonious model, its negative effect (Odds ratio ± SE = −0.009 ± −0.005) on paired-transect dissimilarity was marginal ($X^2 = 3.36$, $p = 0.07$). *Species richness* had a significant effect

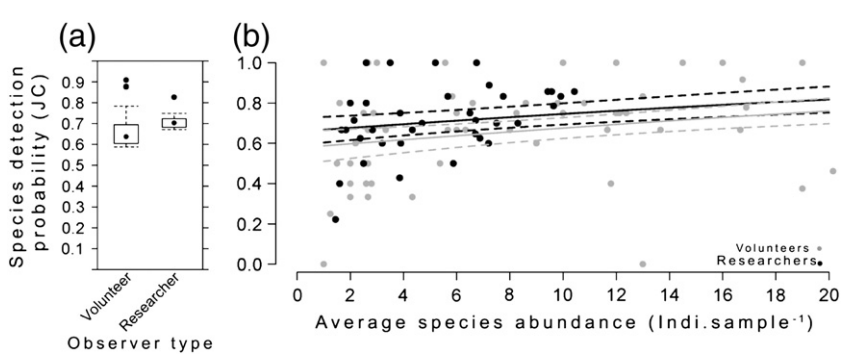

Fig. 2. Predicted species detection probability (*JC*) showing the effect of *Observer type* (a), together with the effect of *Average species abundance* on *JC* (b). The data points in plot (b) are the observed data, while the data for the box and whisker plot (a) and trend lines (b) were predicted from the generalised linear models (±95% confidence intervals) using the observed covariate values.

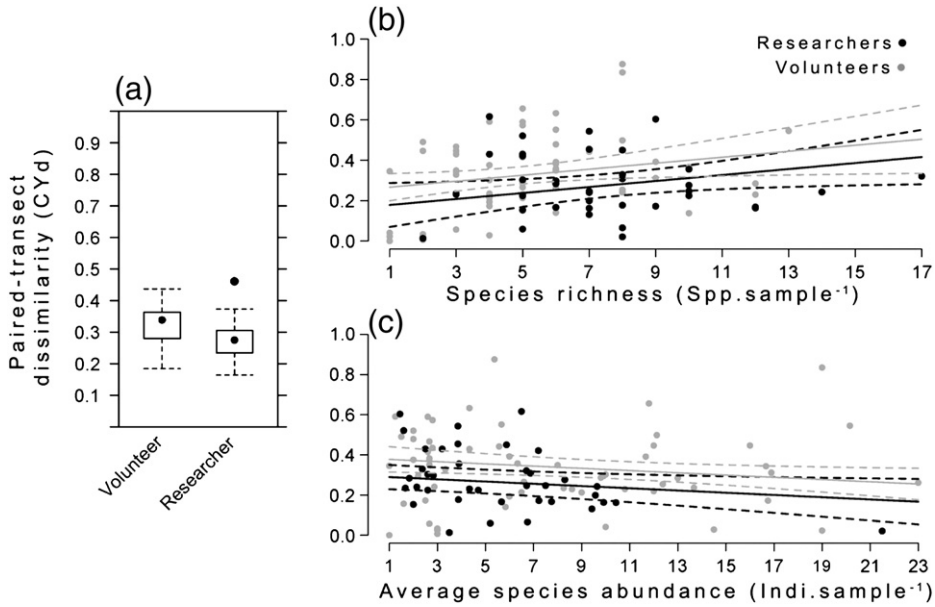

Fig. 3. Predicted paired transect dissimilarity (*CYd*) showing the effect of *Observer type* (a), together with the effect of *Species richness* (b) and *Average species abundance* (c) on *CYd*. The data points in plot (b) and (c) are the observed data, while the data for the box and whisker plot (a) and trend lines (b and c) were predicted from the generalised linear models (±95% confidence intervals) using the observed covariate values.

on paired-transect dissimilarity ($X^2 = 4.82$, $p < 0.05$), with increasing species richness resulting in higher paired-transect dissimilarity (Odds ratio ± SE = 0.015 ± 0.007) (Fig. 3b). An opposite pattern for *Average species abundance* was predicted, with higher abundances resulting in significantly ($X^2 = 4.88$, $p < 0.05$) lower paired-transect dissimilarity scores (Odds ratio ± SE = −0.006 ± 0.003) (Fig. 3c).

3.4. Species analysis

Reef profile was omitted from the most parsimonious models for all the species analysed and its effect on detection probability appeared to vary between species (Supplement 5). *Observer type* was unable to explain any of the variability in the detection probability for roman (*Chrysoblephus laticeps*), hottentot (*Pachymetopon aeneum*), twotone fingerfin (*Chirodactylus brachydactylus*) and barred fingerfin (*Cheilodactylus pixi*), and was dropped from the most parsimonious models (Supplement 5). For all these species, the effect of *Abundance* was significant (Table 3), and able to explain 68.5% of the variability in roman detection probability (Fig. 4a) and 41.6% of the variability in hottentot detection probability (Fig. 4b). The abundance of twotone fingerfin (Fig. 4c) and barred fingerfin (Fig. 4d) significantly influenced the ability of the observers to detect the species (Table 3), however the high variability in the data meant that the models could only explain 23.3% and 10.6% of the variability in the detection probability, respectively.

Table 3
Chi-squared values and significance levels of the covariates included in the most parsimonious binomial generalised linear models investigating the probability that both observers would detect a species if present in a paired-transect.

Species	Abundance	Observer type
Roman	41.59***	–
Hottentot	15.43***	–
Twotone fingerfin	21.65***	–
Barred fingerfin	5.07*	–
Steentjie	26.17***	2.81•
Fransmadam	29.21***	2.3
Redfingers	20.83***	3.08•
Blue hottentot	15.62***	7.62**

Significance level: "***" < 0.001, "**" < 0.01, "*" < 0.05, "•" < 0.1.

Observer type and *Abundance* were included in the most parsimonious models for fransmadam (*Boopsoidea inornata*), steentjie (*Spondyliosoma emarginatum*), blue hottentot (*P. aeneum*) and redfingers (*Cheilodactylus fasciatus*). Higher detection probabilities were predicted for researchers compared to volunteers (Fig. 5), however this effect was only significant for blue hottentot (Table 3). The effect of *Abundance* had a significant positive effect of detection probability for all the above listed species (Table 3). Together *Observer type* and *Abundance* were able to explain 53.0% of the variability in the probability that both observers would detect steentjie (Fig. 5a), 71.5% of the variability in fransmadam (Fig. 5b), 25.5% in the variability of redfingers (Fig. 5c), and 66.9% of the variability in the detection probability that both observers would detect blue hottentot (Fig. 5d) if present in the survey area (Supplement 4). The detection probabilities varied considerably between the different species (Table 4). For example at an abundance of five individuals per transect roman had a 100% detection probability, while at the same abundance barred fingerfin (*C. pixi*) had a detection probability of 64%, and blue hottentot had a detection probability of 95.% and 30% for researchers and volunteers, respectively (Table 4).

4. Discussion

4.1. Validity of assumptions

The direct comparison of the data collected by the two observers conducting a paired-transect relied on the assumptions of independence of data, that only fish within the survey area were counted, that all these fish were equally visible to both observers and that the different species were correctly identified. Because all observers that participated in this experiment were made repeatedly aware that the validity of the research relied on them not assisting each other or sharing information underwater, we assume the first assumption to be valid. The assumption that all fish within the survey area were equally visible to both observers relied on the observers maintaining their positions relative to each other, and on a negligible influence of reef structure on the observers line of sight. During training, divers were assessed on their ability to perform a paired-transect in a pool and all were capable of maintaining their relative positions and buoyancy while swimming side by side down a transect line.

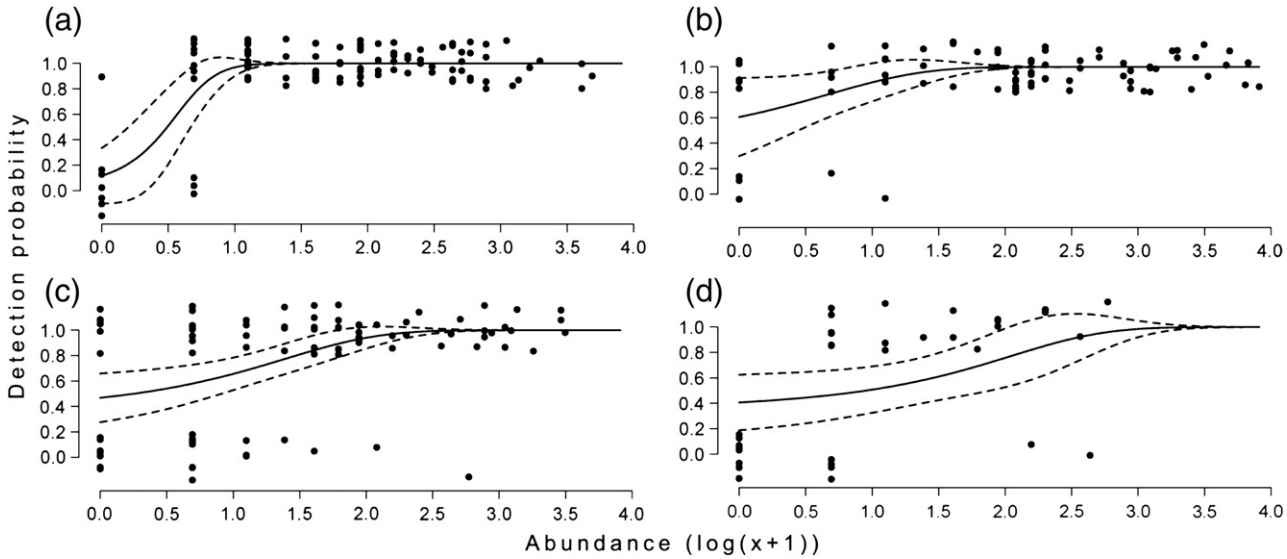

Fig. 4. Predicted probability (±95% confidence intervals) of both observers detecting roman (a), hottentot (b), twotone fingerfin (c), or barred fingerfin (d) during a paired-transect with increasing abundance per species. The x-axis was log transformed to aid graphic presentation and the data points have been jittered to reveal overlapping data.

The models indicated that reef profile had little to no influence on the detection probability or the paired-transect dissimilarity, confirming that reef profile did not bias the paired-transect method.

Knowing which fish to count relies on the ability to estimate distance underwater. The results from this research showed that there was considerable non-systematic variation in the ability of different observers to estimate distance underwater. Inaccuracies in underwater distance estimates have been identified as a source of observer bias in UVC data (Harvey et al., 2004; Thompson and Mapstone, 1997). Biases that are not systematic, be it through space, time, method or observer, have the ability to create misleading conclusions (Edgar et al., 2004). This non-systematic bias observed in the distance estimates is of concern, because it is impossible, or at least very difficult, to correct for, and as such the training should focus on teaching the required skills to standardise and reduce this error between observers. Importantly, during the training tests the distance estimates by observers swimming in pairs were more accurate than estimates from observers swimming alone, suggesting that the presence of a second observer did not negatively influence the paired observers ability to estimate distance.

Within the data there was evidence that volunteers and researchers made species identification errors, where similar looking species were incorrectly identified by one or the other observer in a pair. This suggests that the assumption that all fish were correctly identified might not always be met under field conditions, and that measures need to be put in place to ensure accuracy of identification. Although not considered during this study, an added benefit of paired-transects is that, through consultation between the observers after the survey, obvious identification errors can be determined thereby improving identification skills, and the accuracy of UVC data.

4.2. Observer bias

The data from the paired-transect identified a clear effect of observer type on the accuracy and precision of species detection probability and paired-transect dissimilarity. Although significant differences were

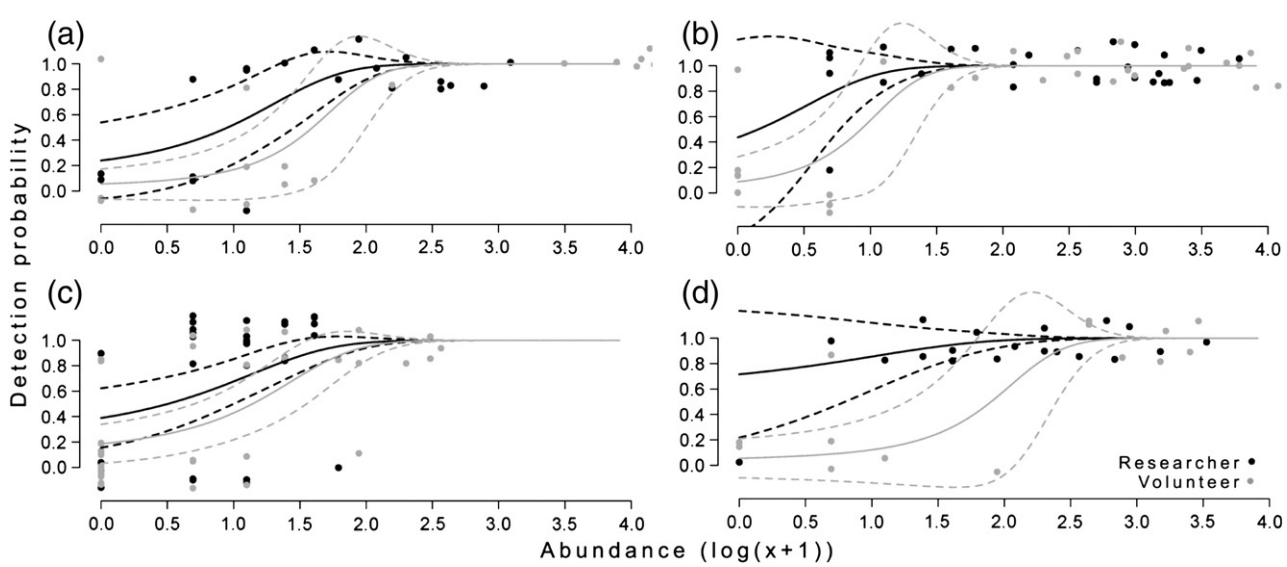

Fig. 5. Predicted probability (±95% confidence intervals) of both researchers (black dots and lines) and both volunteers (grey dots and lines) detecting steentjie (a), fransmadam (b), redfingers(c) or blue hottentot (d) during a paired-transect with increasing abundance per species. The x-axis was log transformed to aid graphic presentation and the data points have been jittered to reveal overlapping data.

Table 4
Predicted detection probabilities (±SE) with increasing abundance for the eight dominant species of fish recorded during the paired-transect surveys. Where observer type was included in the most parsimonious generalised linear models the detection probability is given for researchers and volunteers.

Species	Abundance (indi. sample^{-1})				
	Observer type	1	5	10	20
Roman	Both	0.12 (0.11)	1	1	1
Hottentot	Both	0.61 (0.16)	0.98 (0.03)	1	1
Twotone fingerfin	Both	0.47 (0.1)	0.84 (0.06)	0.98 (0.02)	1
Barred fingerfin	Both	0.41 (0.11)	0.64 (0.1)	0.85 (0.12)	0.98 (0.03)
Steentjie	Researchers	0.24 (0.15)	0.85 (0.12)	1 (0.01)	1
	Volunteers	0.06 (0.06)	0.51 (0.24)	0.97 (0.05)	1
Fransmadam	Researchers	0.44 (0.39)	1 (0.01)	1	1
	Volunteers	0.09 (0.1)	0.98 (0.07)	1	1
Redfingers	Researchers	0.39 (0.12)	0.91 (0.06)	1 (0.01)	1
	Volunteers	0.19 (0.08)	0.78 (0.12)	0.99 (0.01)	1
Blue hottentot	Researchers	0.72 (0.25)	0.95 (0.06)	1 (0.01)	1
	Volunteers	0.06 (0.08)	0.3 (0.24)	0.84 (0.26)	1 (0.01)

detected, the scale of the differences was marginal, particularly for the species detection probability. This was supported in the analysis of the dominant species, where observer type had limited significant influence on the probability that both observers would detect a species if present. However, it is felt that the evidence presented indicates that special care should be taken when involving volunteers in UVC surveys.

This contrasts with the conclusions of past studies that have looked into the reliability of using data collected by volunteers (Darwall and Dulvy, 1996; Edgar and Stuart-Smith, 2009; Mumby et al., 1995). For example, Edgar and Stuart-Smith (2009) compared data collected by volunteers and researchers at the community level by calculating the Bray–Curtis similarity indices between transects performed by volunteers and researchers within the same region at a similar time. They found that variation between individual divers within the different observer groups contributed little to the overall variation between transects. Darwall and Dulvy (1996) arrived at similar conclusions, in that once volunteers had gained experience, the data were of a similar precision to that produced by a researcher.

A number of factors could have contributed to these contradictory results. Firstly, the approach by Edgar and Stuart-Smith (2009) did not take into account instantaneous variation in the fish community, resulting from fish movement, or the response of the fish to the presence of the diver (McClanahan et al., 2007). Instantaneous variation would increase the dissimilarity between separate samples that are not collected in the exact same place and at the exact same time, and in turn increase the variability in the dissimilarity indices blurring the potential effects of observer type, that may have been present in other studies. On the other hand, the approach of Darwall and Dulvy (1996) used a solitary research diver as a control to assess the quality of volunteer data, however they provided no measure of the precision of the data collected by their solitary control diver. The results from this study showed that there was a large amount of variability in the data from both the volunteers and the researchers, with certain researchers consistently associated with data that had higher paired-transect dissimilarity scores than the average for the volunteers. This suggests that it may be inappropriate to assume that all researchers produce data that can be considered to be a baseline.

An alternative explanation may be that training carried out during this study was insufficient compared to other studies (see Darwall and Dulvy, 1996). However, the Frontier Tanzania (Darwall and Dulvy, 1996) programme is conducted in subtropical and tropical environments, which harbour considerably higher diversities of fish species, many of which are small, cryptic and hard to accurately identify. Considering that this study was conducted in temperate waters with lower fish diversity and that small cryptic species were not targeted, it is believed that the duration of the training course was adequate. On the other hand, reef life survey selects only the best volunteers to collect data for the programme (Edgar and Stuart-Smith, 2009). By controlling the quality of observers participating in the programme, the effects of observer type are negated. During this study, all volunteers who passed the tests and attended the pool training sessions were used to collect data in the field. Consequently no quality controls were implemented with respect to the volunteers' field capabilities. This reflected in the results, with certain volunteers consistently associated with samples that had high dissimilarity and low species detection scores. In this regard, the paired-transect method can be useful for detecting observers unable to collect data of a sufficient accuracy and precision.

4.3. Fish community bias

The detection error was greater in paired-transects with high species richness. Lincoln-Smith (1988) and Thompson and Mapstone (1997) observed a similar relationship, where increasing species richness resulted in lower levels of precision in UVC data. The majority of species within a community are rare (Cunningham and Lindenmayer, 2005), suggesting that samples with higher species richness would contain more rare species. Most species recorded during this study were infrequently observed, with 24 of the 32 species occurring in less than 29% of the samples (Supplement 1), and it is possible that these rare species accounted for the increased error in the data due to their low abundance per paired-transect.

Abundance had a marked positive effect on the probability that paired observers would both detect all species along a transect and on the similarity of the counts. When looking only at the dominant species, the effect of abundance was again clear, with the detection probability lower for all species when they were scarce in the survey area (Figs. 2–5). For example, at an abundance of more than ten individuals the detection probability was close to 100% for the dominant species, whereas the detection probability varied between six and 19% for the volunteers and 24 and 72% for the researchers when only one individual of a species was present in a transect. This aligns with current knowledge, where detection probability is higher for more abundant species (Kulbicki, 1998; MacNeil et al., 2008a, b; Thompson and Mapstone, 1997).

There was considerable species specific variation in the detection probability, and in the similarity of the count data suggesting that factors other than abundance may have contributed to this pattern. Detection probability has been shown to increase with the size and behaviour of the fish (Bozec et al., 2011; Kulbicki, 1998). In this study, roman had the highest detection probability despite its moderate abundance. Roman is a large, dominant and conspicuous fish that is indifferent to the presence of divers. In this instance it is likely that these characteristics enhanced the detectability of the species. On the other hand, for the smaller, drab coloured steentjie and blue hottentot, which were abundant in the paired-transect data, a low detection probability was found. It is possible, however, that much of this error was an artefact of misidentification, as the two species are similar in size, behaviour and colouration.

4.4. Accounting for detection probability

The reduced detection probability for rare species will have implications for research investigating the change in fish community structure following disturbance, and comparing fish populations inside and outside of protected areas (MacNeil et al., 2008b). In these instances, as populations of different species grow the detection probability will increase, confounding assessments on the rate and scale of recovery. Often long-term monitoring programmes aim to document population recovery of fishery species. For these objectives

alternative methods that inherently target fishery species, such as controlled angling (Götz et al., 2008) or baited remote underwater video (Willis et al., 2000; Watson et al., 2007), may be better suited. However, some studies suggest that these methods are not as efficient at monitoring entire reef communities as UVC (Colton and Swearer, 2010; Götz et al., 2008), and as such approaches that allow for inclusion of detection probability into abundance estimates from UVC need to be advocated.

Accounting for detection probability is essential for accurate estimation of population abundance (Elphick, 2008; Jenkins and Manly, 2008; Riddle et al., 2010). Methods to estimate detection probability include distance sampling (Kulbicki et al., 2010) and repeated-counts (MacNeil et al., 2008a, b), both of which have received limited use for UVC of reef fish communities. Although these methods offer a means to account for detection probability in UVC, they both come with weaknesses that will influence the precision of the probability estimates. For example, distance-based UVC requires distance to be estimated underwater (Kulbicki and Sarramengna, 1999), which has the potential to introduce non-systematic bias into the data (Harvey et al., 2004). On the other hand, repeated counts rely on CMR models to estimate detectability, however these models assume that the sample is closed (i.e. there is no gain or loss of fish in the time between samples), which is not likely to be the case in open marine environments.

In the terrestrial environment the double-observer method has been applied to estimate visibility bias, observer bias and detection probabilities (Cook and Jacobsen, 1979; Graham and Bell, 1989; Nichols et al., 2000), and if offers a number of advantages over distance sampling and repeated-counts. Firstly, the need to estimate distance underwater is reduced as the observer only needs to estimate the boundary of the survey area. There is no need to capture and mark individuals so that they can be tracked over multiple visits to a sample site, and as the two surveys of the transect area are instantaneous there is no need to return to the sample site. Similarly, as the double-observer method is instantaneous the assumption that the sample population is closed is met.

The dependant double-observer (DDO) (Cook and Jacobsen, 1979; Graham and Bell, 1989) and the IDO, described in the Materials and methods section of this manuscript (Jenkins and Manly, 2008) are the most commonly used double observer techniques. The DDO approach relies on limited, but specific communication between a primary and a secondary observer, and as such is not easily applied underwater, on standard SCUBA equipment. The IDO approach is more suited to underwater applications as no communication is permitted, however the approach requires the location of each fish sighting to be mapped which may unduly complicate UVC surveys of entire fish communities. An alternative approach, that has received limited application, is the unreconciled double-observer method (Riddle et al., 2010). The method is basically an independent double-observer method that does not require matching observations or any communication between observers (Riddle et al., 2010), and is equivalent to the paired-transect method used in this study. The data are analysed with hierarchical models developed for repeated counts of a sample through time (Fiske and Chandler, 2011; Riddle et al., 2010). By using a two stage hierarchical model both the state process that determine abundance or species occurrence at a site, and the detection probability can be accommodated in the abundance estimates (Fiske and Chandler, 2011; Riddle et al., 2010). This was not attempted during this study, due to the failure of assumption four, that all species were correctly identified, and uncertainties around the accuracy of the underwater distance estimates by the volunteers, as this would have led to misleading results. However, by installing mechanisms that ensure correct identification of species as well as accurate estimation of the survey area, the unreconciled double-observer approach may be able to alleviate some of the underlying biases that reduce the diagnostic power of UVC data.

5. Conclusions

One of the benefits of volunteer programmes is the increased manpower that they afford. However, the results from this study suggest that most volunteers produce data that is of a lower quality than that provided by experienced researchers. In this study certain species were easier to survey than others suggesting that volunteer programmes can improve data quality if they focus on a subset of the community that is less prone to observer bias. Similarly, variation between different observers was very high, with certain volunteers consistently producing better data than certain researchers. Volunteer monitoring programmes that only consider the data produced by the best volunteers (e.g. Edgar and Stuart-Smith, 2009) will mitigate some of the observer biases identified in this study, while still raising stakeholder awareness through public participation.

Species abundance was found to have a strong influence on detection probability. Although other factors such as size, behaviour and crypticity may have influenced the data, the ability of the observers to detect species that were scarce in the transect area was poor. The results from this study suggest that it may be feasible to use the unreconciled double-observer method (Riddle et al., 2010), with experienced volunteers or researchers, to correct for detection errors in abundance estimates of subtidal fish populations. For the purpose of this study, the observers would first lay the transect line, and then swim back counting the fish within the transect. While this wouldn't generate bias between observers, it would likely affect the presence of larger species along the transect, and bias the abundance and diversity measures. As such, future research using double-observers to account for detection errors in abundance estimates should either lay the transect out as the observers conduct the survey, or lay it out and provide sufficient time for the fish assemblage to become independent of the initial swim through the area before conducting the survey.

Acknowledgements

Funding for this research was provided by the World Wide Fund for Nature (WWF) South Africa, the Line-Fish Research Division of the Department of Agriculture, Forestry and Fisheries (DAFF), the Elwandle Node of the South African Environmental Observation Network (SAEON), the Mellon Foundation and the British Ecological Society (BES). The authors would like to extend a special gratitude to all the volunteers and researchers who participated in this study. The authors would also like to thank the manuscript reviewers for the constructive comments provided during the publication process. [**ST**]

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.jembe.2013.02.039.

References

Anderson, M.J., Millar, R.B., 2004. Spatial variation and effects of habitat on temperate reef fish assemblages in north-eastern New Zealand. J. Exp. Mar. Biol. Ecol. 305, 191–221.

Bennett, R.H., Götz, A., Sauer, W.H.H., Cowley, P.D., Palmer, R.M., 2009. Optimisation of underwater visual census and controlled angling methods for monitoring subtidal temperate reef fish communities. Afr. J. Mar. Sci. 31, 277–462.

Bernard, A.T.F., Götz, A., 2011. Development of a Volunteer Diver Based Marine Monitoring Programme for the Table Mountain National Park. Technical Report Prepared for WWF South Africa. Stellenbosch, South Africa.

Bozec, Y.-M., Kulbicki, M., Laloe, F., Mou-Tham, G., Gascuel, D., 2011. Factors affecting the detection distances of reef fish: implications for visual counts. Mar. Biol. 158, 969–981.

Cao, Y., Williams, W.P., Bark, A.W., 1997. Similarity measures bias in river benthic Aufwuchs community analysis. Water Environ. Res. 69, 95–106.

Cheal, A.J., Thompson, A.A., 1997. Comparing visual counts of coral reef fish: implications of transect width and species selection. Mar. Ecol. Prog. Ser. 158, 241–248.

Colton, M.A., Swearer, S.E., 2010. A comparison of two survey methods: differences between underwater visual census and baited remote underwater video. Mar. Ecol. Prog. Ser. 400, 19–36.
Cook, R.D., Jacobsen, J.O., 1979. A design for estimating visibility bias in aerial surveys. Biometrics 35, 735–742.
Cunningham, R.B., Lindenmayer, D.B., 2005. Modelling count data of rare species: some statistical issues. Ecology 86, 1135–1142.
Darwall, W.R.T., Dulvy, N.K., 1996. An evaluation of the suitability of non-specialist volunteer researchers for coral reef fish surveys. Mafia Island, Tanzania — a case study. Biol. Conserv. 78, 223–231.
Edgar, G.J., Barrett, N.S., 1999. Effects of the declaration of marine reserves on Tasmanian reef fishes, invertebrates and plants. J. Exp. Mar. Biol. Ecol. 242, 107–144.
Edgar, G.J., Stuart-Smith, R.D., 2009. Ecological effects of marine protected areas on rocky reef communities — a continental-scale analysis. Mar. Ecol. Prog. Ser. 388, 51–62.
Edgar, G.J., Moverley, J., Barrett, N.S., Peters, D., Reed, C., 1997. The conservation-related benefits of a systematic marine biological sampling programme: the Tasmanian reef bioregionalisation as a case study. Biol. Conserv. 79, 227–240.
Edgar, G.J., Barrett, N.S., Morton, A.J., 2004. Biases associated with the use of underwater visual census techniques to quantify the density and size structure of fish populations. J. Exp. Mar. Biol. Ecol. 308, 269–290.
Elphick, C.S., 2008. How you count counts: the importance of methods research in applied ecology. J. Appl. Ecol. 45, 1313–1320.
Fiske, I.J., Chandler, R.B., 2011. unmarked: an R package for fitting hierarchical models of wildlife occurrence and abundance. J. Stat. Softw. 43, 1–23.
Götz, A., Kerwath, S.E., Attwood, C.G., Sauer, W.H.H., 2008. Effects of fishing on population structure and life history of roman *Chrysoblephus laticeps* (Sparidae). Mar. Ecol. Prog. Ser. 362, 245–259.
Graham, A., Bell, R., 1989. Investigating observer bias in aerial survey by simultaneous double-counts. J. Wildl. Manage. 53, 1009–1016.
Halusky, J.G., Seaman, W., Strawbridge, E.W., 1994. Effectiveness of trained volunteer divers in scientific documentation of artificial aquatic habitats. Bull. Mar. Sci. 55, 939–959.
Harborne, A.R., Mumby, P.J., Kappel, C.V., Dahlgren, C.P., Micheli, F., Holmes, K.E., Sanchirico, J.N., Broad, K., Elliot, I.A., Brumbaugh, D.R., 2008. Reserve effects and natural variation in coral reef communities. J. Appl. Ecol. 45, 1010–1018.
Harvey, E., Fletcher, D., Shortis, M.R., Kendrick, G.A., 2004. A comparison of underwater visual distance estimates made by scuba divers and a stereo-video system: implications for underwater visual census of reef fish abundance. Mar. Freshw. Res. 55, 573–580.
Jackson, J.B.C., et al., 2001. Historical overfishing and the recent collapse of coastal ecosystems. Science 293, 629–638.
Jenkins, K.J., Manly, B.F.J., 2008. A double-observer method for reducing bias in faecal pellet surveys of forest ungulates. J. Appl. Ecol. 45, 1339–1348.
Kulbicki, M., 1998. How the acquired behavior of commercial reef fishes may influence the results obtained from visual censuses. J. Exp. Mar. Biol. Ecol. 222, 11–30.
Kulbicki, M., Sarramengna, S., 1999. Comparison of density estimates derived from strip transect and distance sampling for underwater visual censuses: a case study of Chaetodontidae and Pomacanthidae. Aquat. Living Resour. 12, 315–325.
Kulbicki, M., Cornuet, N., Vigliola, L., Wantiez, L., Moutham, G., Chabanet, P., 2010. Counting coral reef fishes: interaction between fish life-history traits and transect design. J. Exp. Mar. Biol. Ecol. 387, 15–23.
Lincoln-Smith, M.P., 1988. Effects of observer swimming speed on sample counts of temperate rocky reef fish assemblages. Mar. Ecol. Prog. Ser. 43, 223–231.
Logan, M., 2010. Biostatistical Design and Analysis Using R: A Practical Guide. Wiley-Blackwell 541.
MacNeil, M.A., Graham, N.A.J., Conroys, M.J., Fonnesback, C.J., Polunin, N.V.C., Rushton, S.P., Chabanet, P., McClanahan, T.R., 2008a. Detection heterogeneity in underwater visual-census data. J. Fish Biol. 73, 1748–1763.
MacNeil, M.A., Tyler, M.H.M., Fonnesbeck, C.J., Rushton, S.P., Polunin, N.V.C., Conroy, M.J., 2008b. Accounting for detectability in reef-fish biodiversity estimates. Mar. Ecol. Prog. Ser. 367, 249–260.
McClanahan, T.R., Graham, N.A.J., Maina, J., Chabanet, P., Bruggemann, J.H., Polunin, N.P.C., 2007. Influence of instantaneous variation on estimates of coral reef fish populations and communities. Mar. Ecol. Prog. Ser. 340, 221–234.
Mumby, P.J., Harborne, A.R., Raines, P.S., Ridley, J.M., 1995. A critical assessment of data derived from Coral Cay conservation volunteers. Bull. Mar. Sci. 56, 737–751.
Nichols, J.D., Hines, J.E., Sauer, J.R., Fallon, F.W., Fallon, J.E., Heglund, P.J., 2000. A double-observer approach for estimating detection probability and abundance from point counts. Auk 117, 393–408.
Pattengill-Semmens, C.V., Semmens, B.X., 2003. Conservation and management applications of the reef volunteer fish monitoring program. Environ. Monit. Assess. 81, 43–50.
R Development Core Team, 2011. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria3-900051-07-0 (URL http://www.R-project.org/).
Riddle, J.D., Pollock, K.H., Simons, T.R., 2010. An unreconciled double-observer method for estimating detection probability and abundance. Auk 127, 841–849.
Thompson, A.A., Mapstone, B.D., 1997. Observer effects and training in underwater visual surveys of reef fishes. Mar. Ecol. Prog. Ser. 154, 53–63.
Thresher, R.E., Gunn, J.S., 1986. Comparative analysis of visual census techniques for highly mobile, reef associated piscivores Carangidae. Environ. Biol. Fishes 17, 93–116.
Vos, P., Meelis, E., Ter Keurs, W.J., 2000. A framework for the design of ecological monitoring programs as a tool for environmental and nature management. Environ. Monit. Assess. 61, 317–344.
Ward-Paige, C., Mills, Flemming J., Lotze, H.K., 2010. Overestimating fish counts by non-instantaneous visual censuses: consequences for population and community descriptions. PLoS One 5 (7), e11722.
Watson, D.L., Harvey, E.S., Kendrick, G.A., Nardi, K., Anderson, M.J., 2007. Protection from fishing alters the species composition of fish assemblages in a temperate-tropical transition zone. Mar. Biol. 152, 1197–1206.
Williams, I.D., Walsh, W.J., Tissot, B.N., Hallacher, L.E., 2006. Impact of observer experience level on counts of fishes in underwater visual surveys. Mar. Ecol. Prog. Ser. 310, 185–191.
Willis, T.J., 2001. Visual census methods underestimate density and diversity of cryptic reef fishes. J. Fish Biol. 59, 1408–1411.
Willis, T.J., Millar, R.B., Babcock, R.C., 2000. Detection of spatial variability in relative density of fishes: comparison of visual census, angling, and baited underwater video. Mar. Ecol. Prog. Ser. 198, 249–260.
Zuur, A.F., Ieno, E.N., Walker, N., Saveliev, A.A., Smith, G.M., 2009. Mixed Effects Models and Extensions in Ecology with R. Springer, New York.
Zuur, A.F., Ieno, E.N., Elphick, C.S., 2010. A protocol for data exploration to avoid common statistical problems. Methods Ecol. Evol. 1, 3–14.